

WGMLEARN - Working Group on Machine Learning in Marine Science

Approved on 18 November 2025

Multiannual

2025/MT/DSTSG02 The Working Group on Machine Learning in Marine Science (WGMLEARN), chaired by Hassan Moustahfid (United States), and Ketil Malde (Norway) will work on ToRs and generate deliverables as listed below.

	Meeting Date(s)	Venue	Report Deadline
2026	To be decided (October)	ICES HQ, Copenhagen, Denmark	29 November 2026
2027	To be decided	To be decided	29 November 2027
2028	To be decided	To be decided	29 November 2028

ToR	Description	Background	Science Plan Codes	Year	Expected Deliverables
a	Develop a list of relevant topics and tasks for the application of Artificial Intelligence (AI) and Machine Learning (ML), with emphasis on unmet challenges specific to the marine domain and potential benefits. Integrate the output from Workshop on Applying AI in ICES (WKAAlI) and link to relevant ICES working groups, research groups, and public datasets.	There is a need to identify gaps in the application of ML in marine sciences to focus future efforts.	4.1, 4.2, 4.3, 4.4, 4.5, 4.6	2026	Review paper (food-for-thought format).
b	Establish contact with other ICES working groups to explore collaboration opportunities and to chart research and knowledge needs. Support the development of aspects of AI at the ICES strategic level.	To prioritize research and dissemination effort, it is necessary to develop and maintain good communications with the users of ML technology.	4.1, 4.2, 4.3, 4.4, 4.5, 4.6	2026, 2027	To prioritize research and dissemination effort, it is necessary to develop and maintain good communications with the users of ML technology.
c	Based on the outcomes of ToR a and b, organize a series of 5–10 events to target important challenges. These can take the form of tutorials, hackathons, workshops, or conference sessions, co-organized with relevant ICES working groups or other entities.	This ToR aims to build capacity within the marine science community on the use of ML and leverage the skills and knowledge of the community. The aim is to bring people together to foster collaboration and innovation.	4.1, 4.2, 4.3, 4.4, 4.5, 4.6	2026, 2027, 2028	Tutorials: Video recordings and reusable learning modules are made available online for community use. Hackathons: Open-source code repositories (e.g. GitHub) containing prototype algorithms, workflows, or data products, and

Each event will lead to a written report.

preparation of scientific or technical peer-reviewed manuscripts.
Workshops:
Workshop reports summarizing discussions, outcomes, and recommendations.

Scientific Justification

The field of machine learning and AI is developing rapidly, and these technologies are and will be instrumental for marine research and resource management. As the technology is explored and established, other ICES expert groups are seeking advice, comments, or other inputs, which WGMLEARN aims to provide. It is also important to identify challenges unique to the marine domain that current mainstream ML research fails to adequately address. Highlighting these challenges and linking to appropriate training sets and test cases will help steer ML research towards high-value opportunities in marine science and management.

Resource Requirements

ICES may help to provide a venue for meetings and workshops at the ICES Headquarters in Copenhagen, Denmark, and the ICES Secretariat can assist with organization and coordination.

Linkages to other ICES Committees or Groups

ACOM, DIG, DSTSG, SCICOM, WGBIOP, WGFAST, WGSMAII